

PROJECT REPORT

INTRODUCTION

This project is regarding Home Automation which was prototyped during the summer of 2021 and refined during the summer of 2023, if you are a student of science, you might find it interesting to go through and learn some new things about IoT (Internet of Things), Cloud Computing and Microcontrollers.

PURPOSE

The very reason for this project was to automate the home appliances (like: Lights, Fans, Air Conditioners. Etc.) and to know the status of every device/appliance right on our mobile devices or laptops or any other device which can access the internet.

APPARATUS

This project required many things from different points in time, here is a list of all the things used during the entirety of this project:

1. ESP8266:



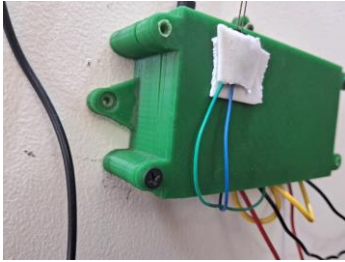
It is a Wi-Fi microchip which means it can connect to any Wi-Fi router/Hotspot within range and I/O devices can exchange relevant signals over the internet. [OBJ] NodeMCU ESP8266 Wi-Fi Module is used in this project. Click [here](#) for the link of the same.

2. 4-Channel Relay Module:



Relay is a device which can connect two wires on command basically acting as a switch, a switch that can be electronically triggered through a digital 0 or 1 signal coming from the microcontroller, here up to 4 appliances can be controlled. Click [here](#).

3. 3D Printed Case:



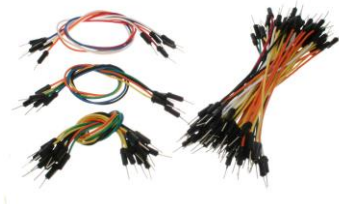
A case was designed using Blender with the intention of holding all the components including the Microcontroller, Relay Module, Wires. Etc. Unfortunately, the data of the same was wiped out of the system for some unforeseen circumstances but a model closest to the same can be found on [Thingiverse](#). Click [here](#).

4. Adapter (with Type B Cable):



An adapter is used to power the ESP and hence the Relay Module. It is important to note that ESP8266 is not 5V tolerant and giving it a direct supply might fry it, however in this project an adapter with almost 4 ~ 4.5V supply was used whereas the security limit (Voltage) for ESP8266 is 3.3V so it may be inferred as it has some tolerance.

5. Wires:



A set of jumper wires were used to make the connections across the relevant pins of the Microcontroller and the Relay Module. Click [here](#) for jumper wires. Also, a set of silver wires were used to connect the retro switches to the Relay Modules.

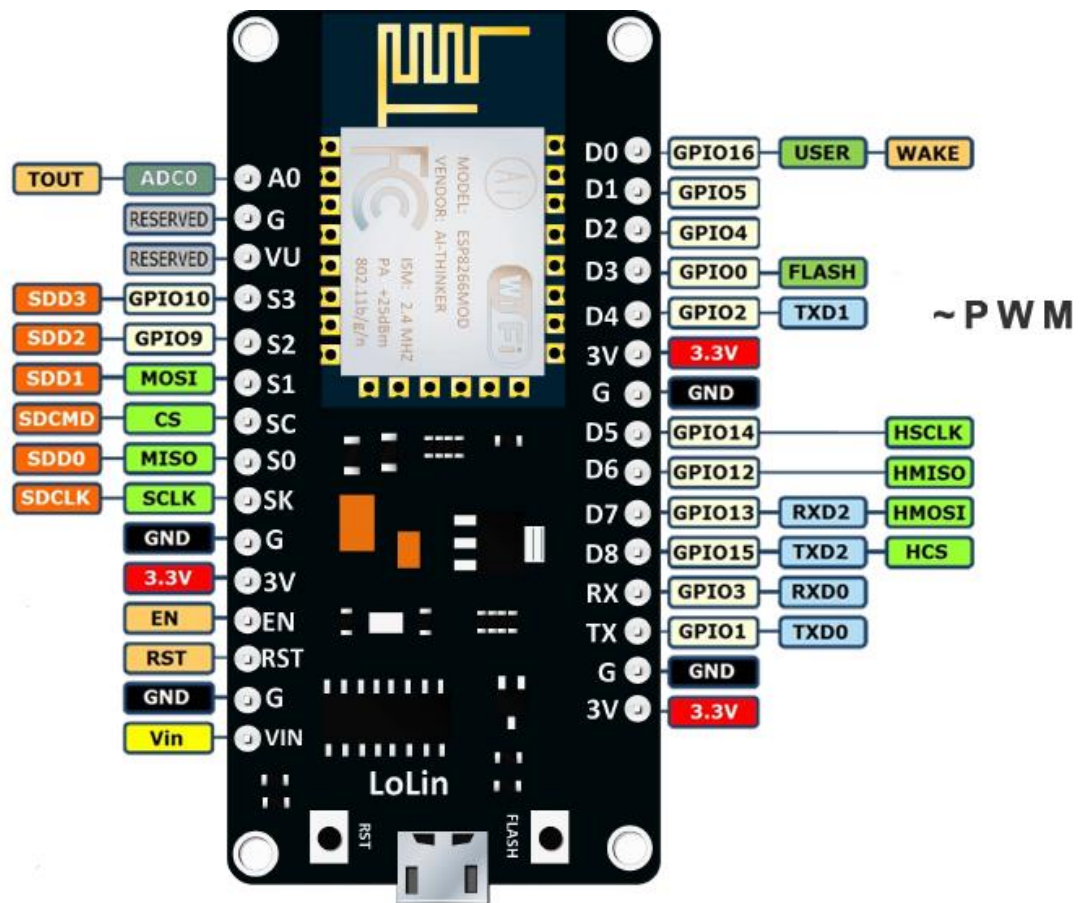
6. IR (Infrared) LED:



An IR (Infrared) LED was used to turn ON and OFF the Air Conditioner.

CIRCUITRY

To understand the circuitry of this project, firstly let us see the pinout diagram of ESP8266:



In this project GPIO (General Purpose Input/Output) Pins 2,5,12,13,14 are used, along with couple of 3.3V power supply and GND ground pins to provide power supply to the Relay Module.

Relay Circuitry:



Image of the Room Automation Project

Note* LED is wired up with ESP8266 using GPIO and GND pin.

Code Snippet

Arduino [IoT Cloud](#) was used to generate control signals to operate the appliances/Things!

Here are some code snippets for this project:

Cloud Variables

ADD

	Name ↓	Last Value	Last Update	
☐	AC CloudSwitch aC;	false	02 Aug 2023 13:38:48	⋮
☐	Balcony_LED CloudSwitch balcony_LED;	false	02 Aug 2023 13:38:33	⋮
☐	Fan CloudSwitch fan;	false	03 Aug 2023 21:47:19	⋮
☐	White_LED CloudSwitch white_LED;	true	02 Aug 2023 13:38:29	⋮
☐	Yellow_LED CloudSwitch yellow_LED;	false	02 Aug 2023 13:38:34	⋮

Living_Room_mar27a.ino

ReadMe.adoc

thingProperties.h

Secret

▼

```
1  /*
2   The following variables are automatically generated and updated when changes are made to the Thing
3
4   CloudSwitch balcony_LED;
5   CloudSwitch fan;
6   CloudSwitch white_LED;
7   CloudSwitch yellow_LED;
8  */
9
10 #include "thingProperties.h"
11 #include <Arduino.h>
12 #include <IRremoteESP8266.h>
13 #include <IRsend.h>
14
15 const uint16_t kIrLed = 4;
16
17 int i = 3;
18 int temp = i;
19
20 IRsend irsend(kIrLed);
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54
55 int yellow = 2;
56 int white = 5;
57 int fan_ = 14;
58 int balcony = 12;
59 int tube = 13;
60
61 void setup() {
62   pinMode(yellow, OUTPUT);
63   pinMode(white, OUTPUT);
64   pinMode(fan_, OUTPUT);
65   pinMode(balcony, OUTPUT);
66   pinMode(tube, OUTPUT);
67   // Initialize serial and wait for port to open:
68   Serial.begin(9600);
69   // This delay gives the chance to wait for a Serial Monitor without blocking if none is found
70   delay(1500);
71
72   // Defined in thingProperties.h
73   initProperties();

```

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73 // Uncomment all change properties
74 initProperties();
75
76 // Connect to Arduino IoT Cloud
77 ArduinoCloud.begin(ArduinoIoTPreferredConnection);
78
79 /*
80  The following function allows you to obtain more information
81  related to the state of network and IoT Cloud connection and errors
82  the higher number the more granular information you'll get.
83  The default is 0 (only errors).
84  Maximum is 4
85 */
86 setDebugLogLevel(2);
87 ArduinoCloud.printDebugInfo();
88
89 irsend.begin();
90 #if ESP8266
91   Serial.begin(115200, SERIAL_8N1, SERIAL_TX_ONLY);
92 #else // ESP8266
93   Serial.begin(115200, SERIAL_8N1);
94 #endif // ESP8266

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```

92   Serial.begin(115200, SERIAL_8N1);
93 #endif // ESP8266
94 }
95
96 void loop() {
97   ArduinoCloud.update();
98 }
99
100 /*
101  Since YellowLED is READ_WRITE variable, onYellowLEDChange() is
102  executed every time a new value is received from IoT Cloud.
103 */
104 void onYellowLEDChange() {
105   if (yellow_LED)
106     digitalWrite(yellow, LOW);
107   else
108     digitalWrite(yellow, HIGH);
109 }
110
111

```

```

116 void onBalconyLEDChange() {
117   if (balcony_LED)
118     digitalWrite(balcony, LOW);
119   else
120     digitalWrite(balcony, HIGH);
121 }
122
123 /*
124  Since Fan is READ_WRITE variable, onFanChange() is
125  executed every time a new value is received from IoT Cloud.
126 */
127 void onFanChange() {
128   if (fan)
129     digitalWrite(fan_, LOW);
130   else
131     digitalWrite(fan_, HIGH);
132 }
133

```

```

133
134 /*
135  Since WhiteLED is READ_WRITE variable, onWhiteLEDChange() is
136  executed every time a new value is received from IoT Cloud.
137 */
138 void onWhiteLEDChange() {
139   if (white_LED)
140     digitalWrite(white, LOW);
141   else
142     digitalWrite(white, HIGH);
143 }

```

```

146 void onACChange() {
147   if (aC)
148   {
149     while (i)
150     {
151       irsend.sendRaw(rawDataON, 211, 38); // Send a raw data capture at 38kHz.
152       delay(500);
153       i = i - 1;
154     }
155     i = temp;
156   }
157   else
158   {
159     while (i)
160     {
161       irsend.sendRaw(rawDataOFF, 211, 38); // Send a raw data capture at 38kHz.
162       delay(500);
163       i = i - 1;
164     }
165     i = temp;
166   }
167 }

```

Challenges

On the boot, some pins of the ESP8266 are HIGH which turns ON the appliances connected to them, and it becomes difficult to handle power cuts due to this fact.

Handling the appliances/Things with the pins which stay LOW on the boot or using a UPS (uninterrupted Power Supply) could be proposed.

References

Arduino IRRemote Library by Ken Sherriff : [GitHub - Arduino-IRremote/Arduino-IRremote: Infrared remote library for Arduino: send and receive infrared signals with multiple protocols](#)